

25.1 Astronomy

Question Paper

Course	CIE A Level Physics
Section	25. Astronomy & Cosmology
Topic	25.1 Astronomy
Difficulty	Hard

Time allowed: 50

Score: /40

Percentage: /100

Question 1a

In recent years, astronomers have discovered sources of fast radio bursts (FRBs) in other galaxies. Studies suggest that these sources may be a type of standard candle.

An FRB source emits intense bursts of radio waves, each burst lasting for a fraction of a second. The closest FRB source is in a massive spiral galaxy 4.60×10^{24} m from the Earth.

A detector of area $1.00 \times 10^{-4} \text{ m}^2$ on the surface of the Earth received bursts of radio waves. In one burst, 9.40×10^{-23} J of energy was received in a time of 1.15 ms.

(i)

Show that the luminosity of the source is about $2 \times 10^{35} \text{ W}$.

[4]

(ii)

When FRB sources were first discovered, some observers suggested that the bursts might be extraterrestrial communications.

Suggest why this is unlikely.

luminosity of the Sun = $3.8 \times 10^{26} \text{ W}$

[2]

[6 marks]

Question 1b

It has been suggested that FRBs are produced by a type of star known as a magnetar.

Magnetars are extremely dense, hot remnants of stars which produce intensely powerful magnetic fields. These magnetic fields power the emission of high-energy radiation, particularly X-rays and gamma rays.

Fig. 1.1 shows how the radiated intensity from a magnetar varies with photon energy.

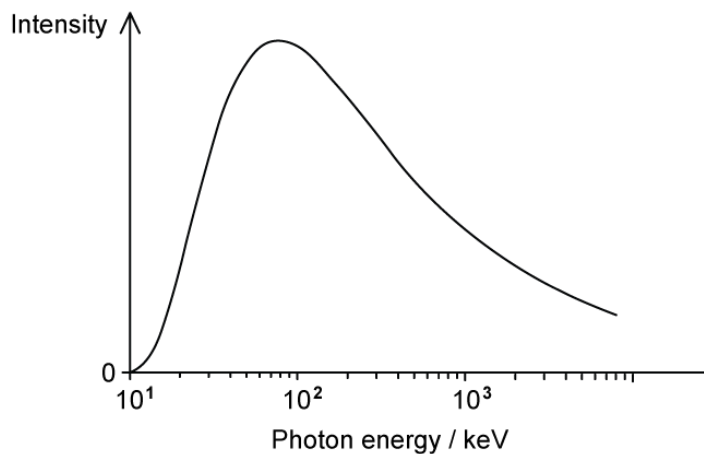


Fig. 1.1

Using Fig. 1.1, determine the surface temperature of the magnetar.

Wien's displacement constant = $2.898 \times 10^{-3} \text{ m K}$

[4 marks]

Question 1c

Some literature about magnetars states they are similar in size to a large city.

Assess the accuracy of this statement.

[4 marks]

Question 2a

Solar panels consisting of combinations of photovoltaic cells use energy in the radiation received from the Sun to generate electricity.

An advertisement for solar panels claims that the intensity of radiation from the Sun incident at the top of the Earth's atmosphere is more than 2 kW m^{-2} .

Assess the validity of this claim.

radius of Sun = $6.96 \times 10^8 \text{ m}$

surface temperature of Sun = 5790 K

distance from Sun to Earth = $1.50 \times 10^{11} \text{ m}$

[4 marks]

Question 2b

A space probe is sent on a mission from Earth to Mars.

Power for the space probe is provided by solar panels that collect radiant energy from the Sun. The solar panels have a total area of 31 m^2 and are directly facing the Sun throughout the journey. The efficiency of the solar panels is 30%.

The mean orbital radius of Mars about the Sun is $2.3 \times 10^{11} \text{ m}$.

Compare the power available to the probe when it leaves Earth's atmosphere to the power available on its arrival at Mars.

[4 marks]

Question 2c

Neptune is the farthest known planet from the Sun in the Solar System. Its distance from the Sun is 30 times greater than the distance of the Earth from the Sun.

A similar space probe to the one in **(b)** is sent on a mission to Neptune.

(i)

Calculate the area of solar panels that would be required to achieve the same availability of power at Neptune as the space probe in **(b)** when it was at Mars.

[3]

(ii)

The International Space Station (ISS) currently holds the record for the largest array of solar panels in space, spanning a total area of 3244 m^2 to supply all of its power.

Using this information, and your answer to **(c)(i)**, comment on the current viability of the mission to Neptune.

[2]

[5 marks]

Question 3a

Fig. 1.1 shows the logarithmic relationship between luminosity $\frac{L}{L_{\odot}}$ and temperature $\frac{T}{T_{\odot}}$ for a sample of stars, where $L_{\odot}$ and $T_{\odot}$ represent the luminosity and temperature of the Sun respectively.

Two stars, X and Y, are shown with their corresponding radii, in terms of the radius of the Sun $R_{\odot}$.

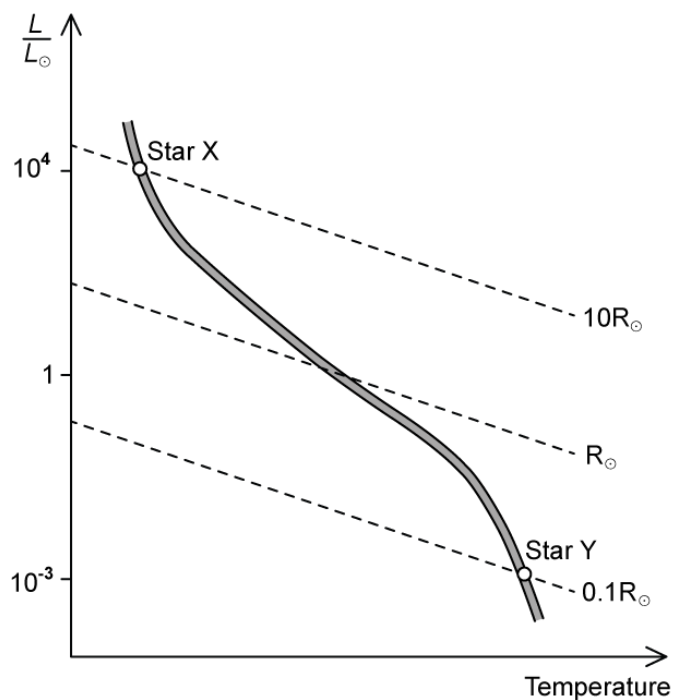


Fig. 1.1

Deduce the colours of star X and star Y using the information in Fig. 1.1 and Table 1.1.

Table 1.1

star colour	surface temperature / $T_{\odot}$
blue	> 5.5
blue-white	$1.7 - 5.5$
white	$1.3 - 1.7$
yellow-white	$1.0 - 1.3$
yellow	$0.8 - 1.0$
orange	$0.6 - 0.8$
red	< 0.6

[4 marks]

Question 3b

The relationship between the mass M and luminosity L of the stars shown on Fig. 1.1 is given by

$$L \propto M^{3.5}$$

Using this mass-luminosity relation and information from Fig. 1.1, determine the ratio

$$\frac{\text{density of star } X}{\text{density of star } Y}$$

[3 marks]

Question 3c

In 2017, an ultra-cool star in our galaxy Trappist-1 was discovered with at least five of its own orbiting planets.

Astronomers have studied it extensively to determine if there is a possibility of finding life on some of the planets orbiting Trappist-1.

Table 1.1 below shows some data.

Table 1.1

	Trappist-1	Sun
Luminosity $L / L_{\odot}$	5.26×10^{-4}	1
Surface temperature $T / T_{\odot}$	0.4	1
Radius of star $/ R_{\odot}$	R	1
Distance between Earth and Sun / m	–	1.5×10^{11}
Distance between planets and Trappist-1 / m	1.6×10^9 to 9.0×10^9	–

The temperature T , in K, of a planet, its distance d from the star and the luminosity L of the star are related by the expression

$$\frac{T^4 d^2}{L} = 419 \times 10^3$$

(i)

Show that the radius R of Trappist-1 is smaller than the radius of the Sun.

[2]

(ii)

The average temperature of the Earth is about 290 K.

Using data from Table 1.1, comment on the possibility of life existing on some of the planets orbiting Trappist-1.

[4]

[6 marks]

Model Answers: Hard

1a

(a)

(i) Show that the luminosity of the source is about $2 \times 10^{35} \text{ W}$:

List the known quantities:

- Distance from Earth, $d = 4.60 \times 10^{24} \text{ m}$
- Area of detector, $A = 1.00 \times 10^{-4} \text{ m}^2$
- Energy of burst, $E = 9.40 \times 10^{-23} \text{ J}$
- Time of burst, $t = 1.15 \text{ ms} = 1.15 \times 10^{-3} \text{ s}$

Determine the power of the burst of radio waves:

- Power: $P = \frac{E}{t} = \frac{9.40 \times 10^{-23}}{1.15 \times 10^{-3}} = 8.17 \times 10^{-20} \text{ W}$ [1 mark]

Determine the intensity of the burst:

- Intensity: $F = \frac{P}{A} = \frac{8.17 \times 10^{-20}}{1.00 \times 10^{-4}} = 8.17 \times 10^{-16} \text{ W m}^{-2}$ [1 mark]

Use the inverse square law of flux equation to find luminosity:

- Inverse square law of flux: $F = \frac{L}{4\pi d^2}$
- $L = 4\pi d^2 F = 4\pi \times (4.60 \times 10^{24})^2 \times (8.17 \times 10^{-16})$ [1 mark]
- $L = 2.17 \times 10^{35} = 2.2 \times 10^{35} \text{ W}$ [1 mark]

(ii) It is unlikely that FRB sources are extraterrestrial communications because:

- The luminosity of the FRB source is much larger than the luminosity of the Sun; [1

mark]

- A power output this large is unlikely to be artificially produced; [1 mark]

[Total: 6 marks]

The marks for part (ii) can only be gained if the answer for part (i) is significantly larger than the value given for the Sun's luminosity

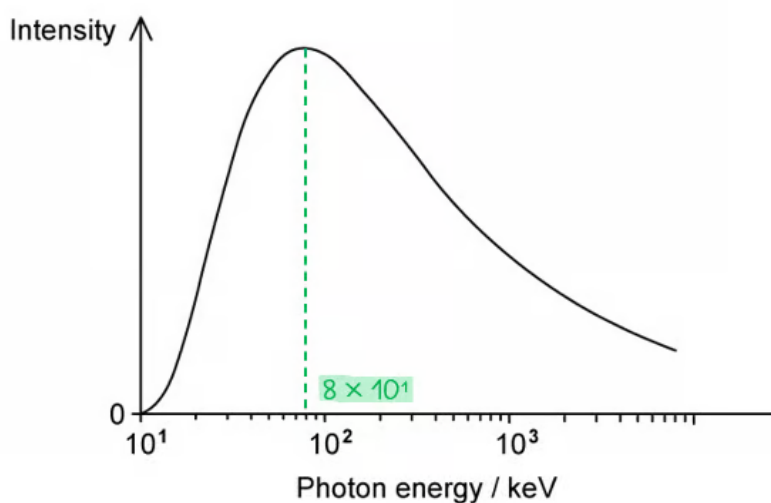
1b

(b) Determine the surface temperature of the magnetar:

List the known quantities:

- Wien's displacement constant = $2.898 \times 10^{-3} \text{ m K}$
- Planck's constant, $h = 6.63 \times 10^{-34} \text{ J s}$
- Speed of light, $c = 3.0 \times 10^8 \text{ m s}^{-1}$
- $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$

Use the graph to obtain a value for the maximum energy:



- (From the graph) maximum energy, $E_{\max} = 8 \times 10^1 = 80 \text{ keV}$ [1 mark]

Acceptable range for maximum energy = 75 to 80 keV

Convert the maximum energy into the maximum wavelength:

- Photon energy: $E = \frac{hc}{\lambda} \Rightarrow \lambda_{\max} = \frac{hc}{E_{\max}}$

- $\lambda_{\max} = \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{(80 \times 10^3)(1.6 \times 10^{-19})}$
- $\lambda_{\max} = 1.554 \times 10^{-11} \text{ m}$ [1 mark]

Acceptable range for max. wavelength = $(1.55 \text{ to } 1.66) \times 10^{-11} \text{ m}$

Use Wien's Law to calculate the temperature of the magnetar:

- Wien's law: $T \propto \frac{1}{\lambda_{\max}} \Rightarrow \lambda_{\max} T = 2.898 \times 10^{-3} \text{ m K}$
- $T = \frac{2.898 \times 10^{-3}}{\lambda_{\max}} = \frac{2.898 \times 10^{-3}}{1.554 \times 10^{-11}}$ [1 mark]
- $T = 1.865 \times 10^8 = 1.9 \times 10^8 \text{ K}$ [1 mark]

Acceptable range for surface temperature = $(1.7 \text{ to } 1.9) \times 10^8 \text{ K}$

[Total: 4 marks]

1c

(c) Determine the diameter of the magnetar and compare it to the size of a city:

List the known quantities:

- Luminosity of FRB/magnetar, $L = 2.17 \times 10^{35} \text{ W}$ (from **(a)**)
- Surface temperature of the magnetar, $T = 1.865 \times 10^8 \text{ K}$ (from **(b)**)
- Stefan-Boltzmann constant, $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Use the Stefan-Boltzmann Law to calculate the radius of the magnetar:

- Stefan-Boltzmann Law: $L = 4\pi R^2 \sigma T^4 \Rightarrow R = \sqrt{\frac{L}{4\pi \sigma T^4}}$
- Radius: $R = \sqrt{\frac{2.17 \times 10^{35}}{4\pi \times (5.67 \times 10^{-8}) \times (1.865 \times 10^8)^4}}$ [1 mark]
- Radius: $R = 15\,866 = 15.9 \text{ km}$ [1 mark]

Estimate the size of a city and compare it to the diameter of the magnetar:

- Diameter of the magnetar = $2 \times 15.9 = 31.8 \text{ km}$
- A reasonable estimate for the size of a large city is 10^1 to 10^2 km ; [1 mark]
- Hence the statement from the literature is correct; [1 mark]

[Total: 4 marks]

2a

(a) Determine the radiant flux intensity at the top of the Earth's atmosphere:

List the known quantities:

- Radius of Sun, $r = 6.96 \times 10^8 \text{ m}$
- Surface temperature of Sun, $T = 5790 \text{ K}$
- Distance from Sun to Earth, $d = 1.50 \times 10^{11} \text{ m}$
- Stefan-Boltzmann constant, $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Calculate the luminosity of the Sun:

- Stefan-Boltzmann law: $L = 4\pi r^2 \sigma T^4$
- $L = (5.67 \times 10^{-8}) \times 4\pi \times (6.96 \times 10^8)^2 \times (5790)^4$
- $L = 3.879 \times 10^{26} \text{ W}$ [1 mark]

Calculate the radiant flux intensity:

- Flux intensity: $F = \frac{L}{4\pi d^2}$
- $F = \frac{3.879 \times 10^{26}}{4\pi \times (1.50 \times 10^{11})^2} = 1371.92 \text{ W m}^{-2}$ [1 mark]
- $F = 1.37 \text{ kW m}^{-2}$ [1 mark]

Conclusion:

- $1.37 \text{ kW m}^{-2} < 2 \text{ kW m}^{-2}$ so the claim is false; [1 mark]

- $1.37 \text{ kW m}^{-2} < 2 \text{ kW m}^{-2}$ so the claim is false; [1 mark]

[Total: 4 marks]

'Assess the validity of this claim' normally means 'check this claim is correct'

You can do this by calculating the value for the intensity, and then seeing how that value compares with the value they have mentioned

You must **always** write a conclusion statement stating whether the claim is valid or not **and why**

This is best done quantitatively i.e. using numbers ($X < Y$ so Y is not valid)

2b

(b) Determine the power available to the probe at Earth and Mars and compare:

List the known quantities:

- Radiant flux intensity at Earth, $F = 1370 \text{ W m}^{-2}$ (from **(a)**)
- Surface area of solar panels, $A = 31 \text{ m}^2$
- Efficiency of solar panels = 30% = 0.3
- Distance from Sun to Earth, $d_E = 1.5 \times 10^{11} \text{ m}$
- Distance from Sun to Mars, $d_M = 2.3 \times 10^{11} \text{ m}$

Determine the power available to the probe at Earth:

- Power: $P = FA$
- The panels have an efficiency of 30%, so
- Total power available (at Earth): $P_E = 0.3 \times 1370 \times 31 = 12\,740 \text{ W}$ [1 mark]

Determine the power available to the probe at Mars:

- Radiant flux intensity: $F = \frac{L}{4\pi d^2}$
- Since $F \propto \frac{1}{d^2}$ and $P \propto F \Rightarrow P \propto \frac{1}{d^2}$

- Ratio: $\frac{P_M}{P_E} = \left(\frac{d_E}{d_M}\right)^2 \Rightarrow P_M = P_E \times \left(\frac{d_E}{d_M}\right)^2$
- Total power available (at Mars): $P_M = 12\,740 \times \left(\frac{1.5 \times 10^{11}}{2.3 \times 10^{11}}\right)^2$ [1 mark]
- Total power available (at Mars): $P_M = 5420 \text{ W}$ [1 mark]

Conclusion:

- The power available to the probe at Mars is 5420 W compared to 12 740 W at Earth, this is a decrease of 42.5%; [1 mark]

[Total: 4 marks]

2c

(c)

(i) Calculate the area of solar panels required to reach Neptune:

List the known quantities:

- Radiant flux intensity at Earth, $F_E = 1370 \text{ W m}^{-2}$ (from (a))
- Distance from Sun to Earth = d_E
- Distance from Sun to Neptune, $d_M = 30d_E$
- Required power at Neptune, $P = 5420 \text{ W}$ (from (b))

Determine the radiant flux intensity at Neptune:

- Radiant flux intensity: $F = \frac{L}{4\pi d^2}$
- Ratio: $\frac{F_N}{F_E} = \left(\frac{d_E}{d_N}\right)^2 \Rightarrow F_N = F_E \times \left(\frac{d_E}{d_N}\right)^2$
- Radiant flux intensity at Neptune: $F_N = 1370 \times \left(\frac{d_E}{30d_E}\right)^2 = 1370 \times \left(\frac{1}{30}\right)^2$ [1 mark]

- Radiant flux intensity at Neptune: $F_N = 1.52 \text{ W m}^{-2}$ [1 mark]

Determine the required area of the panels:

- Power: $P = FA \Rightarrow A = \frac{P}{F}$
- Area of panels required: $A = \frac{5420}{1.52} = 3570 \text{ m}^2$ [1 mark]

(ii) Comment on the viability of the mission:

- Since $3570 \text{ m}^2 > 3244 \text{ m}^2$, this is not (currently) a viable mission; [1 mark]

Any **one** from:

- The ISS stays at a constant distance from the Sun whereas the probe will be moving away (from the Sun); [1 mark]
- The mass of the solar panels would make the launch (from Earth) difficult; [1 mark]
- A different energy source (e.g. nuclear power) would be more efficient / compact; [1 mark]
- The energy needs of the ISS and space probe are different; [1 mark]

[Total: 5 marks]

3a

(a) Deduce the colours of X and Y:

List the known quantities:

- Luminosity of X, $L_X = 10^4 L_\odot$
- Luminosity of Y, $L_Y = 10^{-3} L_\odot$
- Radius of X, $R_X = 10 R_\odot$
- Radius of Y, $R_Y = 0.1 R_\odot$

Use the Stefan-Boltzmann law to determine the temperatures of X and Y:

- Stefan-Boltzmann law: $L = 4\pi R^2 \sigma T^4 \Rightarrow T \propto \sqrt[4]{\frac{L}{R^2}}$
- Temperature of the Sun: $T_{\odot} \propto \sqrt[4]{\frac{L_{\odot}}{(R_{\odot})^2}}$
- Temperature of star X: $T_X \propto \sqrt[4]{\frac{10^4 L_{\odot}}{(10R_{\odot})^2}}$
- Temperature of star X: $T_X \propto \sqrt[4]{\frac{10^4}{10^2}} \sqrt[4]{\frac{L_{\odot}}{(R_{\odot})^2}} = 3.16 T_{\odot}$ [1 mark]
- Temperature of star Y: $T_Y \propto \sqrt[4]{\frac{10^{-3} L_{\odot}}{(0.1R_{\odot})^2}}$
- Temperature of star Y: $T_Y \propto \sqrt[4]{\frac{10^{-3}}{0.1^2}} \sqrt[4]{\frac{L_{\odot}}{(R_{\odot})^2}} = 0.56 T_{\odot}$ [1 mark]

Identify the colours of X and Y:

- Since $T_X = 3.16 T_{\odot}$, star X is blue-white; [1 mark]
- Since $T_Y = 0.56 T_{\odot}$, star Y is red; [1 mark]

[Total: 4 marks]

3b

(b) Determine the ratio of densities of X and Y:

List the known quantities:

- Luminosity of X, $L_X = 10^4 L_{\odot}$
- Luminosity of Y, $L_Y = 10^{-3} L_{\odot}$
- Radius of X, $R_X = 10R_{\odot}$
- Radius of Y, $R_Y = 0.1R_{\odot}$

Determine the ratio of the masses:

- Mass-luminosity relation: $L \propto M^{3.5}$
- Ratio of masses: $\frac{L_X}{L_Y} = \left(\frac{M_X}{M_Y}\right)^{3.5} \Rightarrow \frac{M_X}{M_Y} = \left(\frac{L_X}{L_Y}\right)^{\frac{1}{3.5}}$
- Ratio of masses: $\frac{M_X}{M_Y} = \left(\frac{10^4}{10^{-3}}\right)^{\frac{1}{3.5}} = 10^2$ [1 mark]

Determine the ratio of the volumes:

- Volume of a sphere: $V \propto R^3$
- Ratio of volumes: $\frac{V_X}{V_Y} = \left(\frac{R_X}{R_Y}\right)^3$
- Ratio of volumes: $\frac{V_X}{V_Y} = \left(\frac{10}{0.1}\right)^3 = 10^6$ [1 mark]

Determine the ratio of the densities:

- Density: $\rho = \frac{M}{V}$
- Ratio of densities: $\frac{\rho_X}{\rho_Y} = \left(\frac{M_X}{M_Y}\right) \div \left(\frac{V_X}{V_Y}\right)$
- Ratio of densities: $\frac{\rho_X}{\rho_Y} = \frac{10^2}{10^6} = 10^{-4}$ [1 mark]

[Total: 3 marks]

3c

(c)

(i) Show that the radius R of T-1 is smaller than the radius of the Sun

List the known quantities:

- Luminosity of T-1, $L = (5.26 \times 10^{-4})L_{\odot}$
- Surface temperature of T-1, $T = 0.4T_{\odot}$
- Radius of T-1 = R

Use the Stefan-Boltzmann law to determine the radius of T-1:

- Stefan-Boltzmann law: $L = 4\pi r^2 \sigma T^4 \Rightarrow R = \sqrt{\frac{L}{4\pi\sigma T^4}}$
- Radius of the Sun: $R_{\odot} = \sqrt{\frac{L_{\odot}}{4\pi\sigma(T_{\odot})^4}}$
- Radius of T-1: $R = \sqrt{\frac{(5.26 \times 10^{-4})L_{\odot}}{4\pi\sigma(0.4T_{\odot})^4}}$
- Radius of T-1: $R = \sqrt{\frac{5.26 \times 10^{-4}}{0.4^4}} \times R_{\odot} = 0.14 R_{\odot}$ [1 mark]

Conclusion:

- $0.14 < 1$, therefore, the radius of T-1 is smaller than the radius of the Sun; [1 mark]

(ii) Consider whether life may be possible on some of the planets orbiting Trappist-1:

List the known quantities:

- Earth-Sun distance = 1.5×10^{11} m
- Minimum T1-planet distance = 1.6×10^9 m
- Maximum T1-planet distance = 9.0×10^9 m
- Temperature of Earth = 290 K

Determine the luminosity of T-1:

- (From question): $\frac{T^4 d^2}{L} = 419 \times 10^3 \Rightarrow L = \frac{T^4 d^2}{419 \times 10^3}$
- Luminosity of the Sun: $L_{\odot} = \frac{290^4 \times (1.5 \times 10^{11})^2}{419 \times 10^3} = 3.8 \times 10^{26} \text{ W}$
- Luminosity of T-1: $L = (5.26 \times 10^{-4}) \times (3.8 \times 10^{26}) = 2.0 \times 10^{23} \text{ W}$ [1 mark]

Determine the range of temperatures of planets around T-1:

- $\frac{T^4 d^2}{L} = 419 \times 10^3 \Rightarrow T = \sqrt[4]{\frac{L \times (419 \times 10^3)}{d^2}}$
- Inner-most planet temperature: $T_{min} = \sqrt[4]{\frac{(419 \times 10^3) \times (2.0 \times 10^{23})}{(9.0 \times 10^9)^2}} = 179 \text{ K}$
- Outer-most planet temperature: $T_{max} = \sqrt[4]{\frac{(419 \times 10^3) \times (2.0 \times 10^{23})}{(1.6 \times 10^9)^2}} = 425 \text{ K}$
- $T_{min} = 180 \text{ K}$ **AND** $T_{max} = 430 \text{ K}$ [1 mark]

Write a conclusion:

- With temperatures ranging from 179 to 425 K it is very likely that life could exist on a planet around Trappist-1; [1 mark]

And max **one** from:

- Trappist-1 is smaller / cooler than the Sun; [1 mark]
- The planets orbit closer to Trappist-1; [1 mark]
- It is possible for the temperature on the planets to be similar to Earth; [1 mark]
- For life to exist, temperature is not the only factor; [1 mark]

[Total: 6 marks]